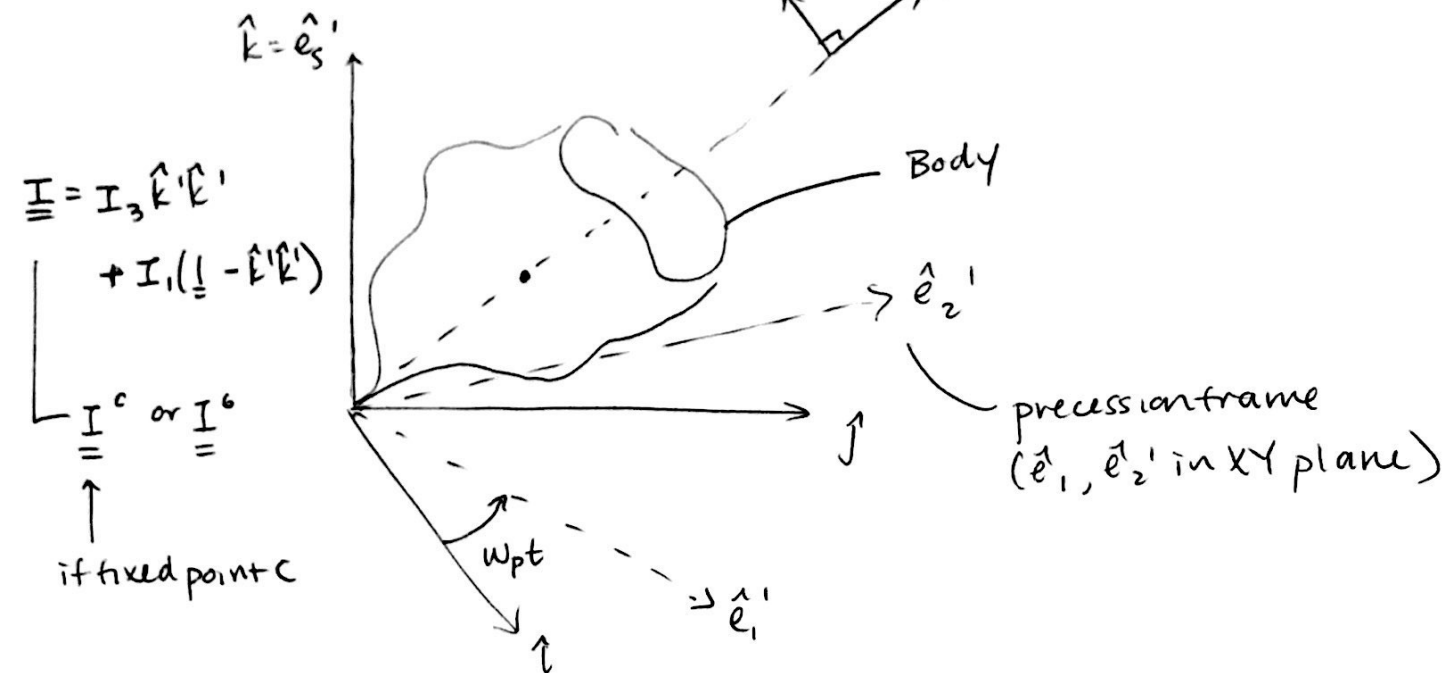


Today:
- steady precession

$$\vec{\omega} = \omega_p \hat{k} + \omega_s \hat{k}'$$

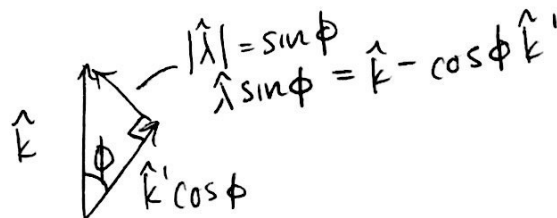
$\nwarrow \quad \nearrow$
 ω_{tor} constants



$$\begin{aligned} \vec{H} &= \underline{I} \cdot \vec{\omega} \\ &= (I_3 \hat{k}'\hat{k}' + I_1 (\underline{1} - \hat{k}'\hat{k}')) \cdot (\omega_p \hat{k} + \omega_s \hat{k}') \\ &= \leftarrow \text{last class} \\ &= \omega_p I_1 \hat{k} + ((I_3 - I_1) \cos \phi \omega_p + I_3 \omega_s) \hat{k}' \quad (*) \end{aligned}$$

Alternative derivation:

$$\vec{\omega} = \omega_p \hat{k} + \omega_s \hat{k}'$$



$$\vec{\omega} = \omega_s \hat{k}' + \omega_p \hat{k}' \cos \phi + \omega_p (\hat{k} - \cos \phi \hat{k}') \underbrace{\sin \phi}_{\hat{k} \sin \phi}$$

$$\vec{H} = \omega_s \hat{k}' I_3 + \omega_p \hat{k}' \cos \phi I_3 + \omega_p (\hat{k} - \hat{k}' \cos \phi) I_1$$

check w/ (*) — ✓

For dynamics need $\dot{\vec{H}}$
L/c or l/b

$\vec{H}$ precesses w/ precessing frame

$$\dot{\vec{H}} = \vec{\omega}_p \times \vec{H}$$

$\omega_p \hat{k}$

$$\begin{aligned} \dot{\vec{H}} &= \omega_p \hat{k} \times (\quad (*) \quad) \\ &= \omega_p ((\text{no } \hat{k} \text{ terms}) *) \sin \phi \hat{e}_z' \end{aligned}$$

$$\boxed{\dot{\vec{H}} = ((I_3 - I_1) \cos \phi \omega_p + I_3 \omega_s) \sin \phi \omega_p \hat{e}_z'}$$

Free precession:

$$\vec{M}_{/b}' = \dot{\vec{H}}_{/b}$$

$$\Rightarrow \vec{O} = \dot{\vec{H}}_{/b}$$

$$\Rightarrow \dot{\vec{H}}_{/b} \parallel \hat{k}$$

$$\Rightarrow (I_3 - I_1) \cos \phi \omega_p + I_3 \omega_s = 0$$

• applies to Feynman plate, football, etc

Feynman Plate:

$$\vec{\omega}_{\text{tot}} = \omega_s \hat{k}' + \omega_p \hat{k}$$

$$\approx (\omega_s + \omega_p) \hat{k}$$

$\phi \ll 1$, $\hat{k}$ almost $\parallel$ to $\hat{k}'$
plate: $I_3 = 2I_1$

$$\Rightarrow \omega_p + 2\omega_s = 0$$

$$\omega_{\text{tot}} = -\frac{\omega_p}{2} + \omega_p$$

$\omega_{\text{tot}} = \frac{\omega_p}{2}$

"wobbles twice as fast as it spins"